

Cooling System Design for FSAE Electric Vehicle

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Abstract

This internal report establishes an evidence-led thermal-management baseline for a rear-drive electric Formula Student powertrain comprising one AMK KW26-S5-FSE-4Q inverter and two DYNASYN DD5-14-10 motors. Revision 12 remains restricted to reproducible numerical verification and evidence closure: it makes no procurement, hardware-release or physical-validation claim. Candidate hardware is illustrated with its evidence status and consolidated in a register; no image or register row implies selection. Coupled steady and finite-volume coolant models reject the passive Boyd/SPAL/CWA E3 model at 40°C air; the finite-volume model closes its semidiscrete energy balance to 3.2×10^{-12} W and passes the 0.1 K mesh/time-step gate. E7 establishes a numerically plausible active two-temperature envelope. The public-data E8 steady model passes its declared baseline, but bisection resolves narrow local limits: 68.0168% condenser-fan command, 43.5217% DD5-fan command, 67.0258% common B3 usable UA , 1.2600 bar KW26-loop loss and 3.134 kW two-motor DD5 coolant heat. Five deterministic interaction grids show that these limits cannot be combined independently. The 15 August public-primary extension verifies component identities and applicability limits but finds no application KW26 loss, DD5 jacket-heat partition, exact CWA150 performance curve, condition-complete B3 duty or target-vehicle waveform; therefore no numerical input or procurement status is promoted. Fifty-nine reduced-order/admission and twenty-one OpenFOAM method-qualification tests pass. A remote OpenFOAM 14 four-level surrogate air-path campaign passes mesh integrity, conservation, settled-state, fan-curve-domain and 2% finest-pair refinement gates: fine-to-extra-fine mass flow and pressure loss change by 0.959% and 0.0145%. This qualifies only the idealized core-matched numerical method. Controlled supplier data, accepted bench/vehicle measurements, installed CAD and measured boundary conditions remain unavailable.

Keywords: Formula SAE, Electric Vehicle, Cooling System, Radiator, Fan, Pump, Catch Can, Swirl Pot, Hoses, FSAE Rules

Contents

1	Cooling-System Topology: E3 Rejected, E8 Numerical Model	4
1.1	Active two-temperature architecture	4
2	Competition Compliance Baseline	5
3	Powertrain Cooling Requirements and Data Status	6
3.1	Verified manufacturer limits	7
3.2	Heat-load model: unresolved inputs, not fixed facts	7
4	Evidence-Limited Thermal and Hydraulic Baseline	8
4.1	Purpose and boundary	8
4.2	Minimum liquid-side capacity	8
4.3	Strict high-ambient design condition	8
4.4	Governing theory and falsifiable predictions	9
4.5	Coupled numerical result, not physical validation	10
4.6	Finite-volume transient heat-transfer verification	11
4.7	E7 active two-temperature assumption screen	11
4.8	E8 public-data numerical model	12
4.9	Remaining evidence limits	13
5	Radiator Numerical Screening: E3 Failure and E8 Reference	14
6	Fan and Air-Path Numerical Screening	15
7	Pump and Hydraulic Numerical Screening	16
8	Overflow Containment and Catch Can	18
9	Degassing and Header-Tank Strategy	18
10	Coolant Chemistry, Hoses and Fittings	19
11	Future Physical Validation Plan	20
12	Executed Numerical Verification	21
12.1	Scope, provenance and reproduction	21
12.2	Public-evidence closure boundary	22
12.3	E3 independent verification	22

12.4 E7 and E8 baseline results	23
12.5 Resolved E8 boundaries	24
12.6 E8 interaction domains	25
12.7 OpenFOAM status and excluded physics	26
12.8 Numerical decision	27
13 Consolidated Candidate Hardware Register	27

1 Cooling-System Topology: E3 Rejected, E8 Numerical Model

The original single-loop E3 topology is retained only as a rejected numerical baseline. Figure 1 shows the E8 logical simulation reference: separate KW26 and DD5 coolant loops, an R134a cycle and a third coolant loop between the refrigerant condensers and ambient-air cores. It is a numerical network definition, not an installation or hardware-release drawing. Hose lengths, fittings, fill/bleed points, relief discharge, mounts and local duct geometry remain unavailable inputs.

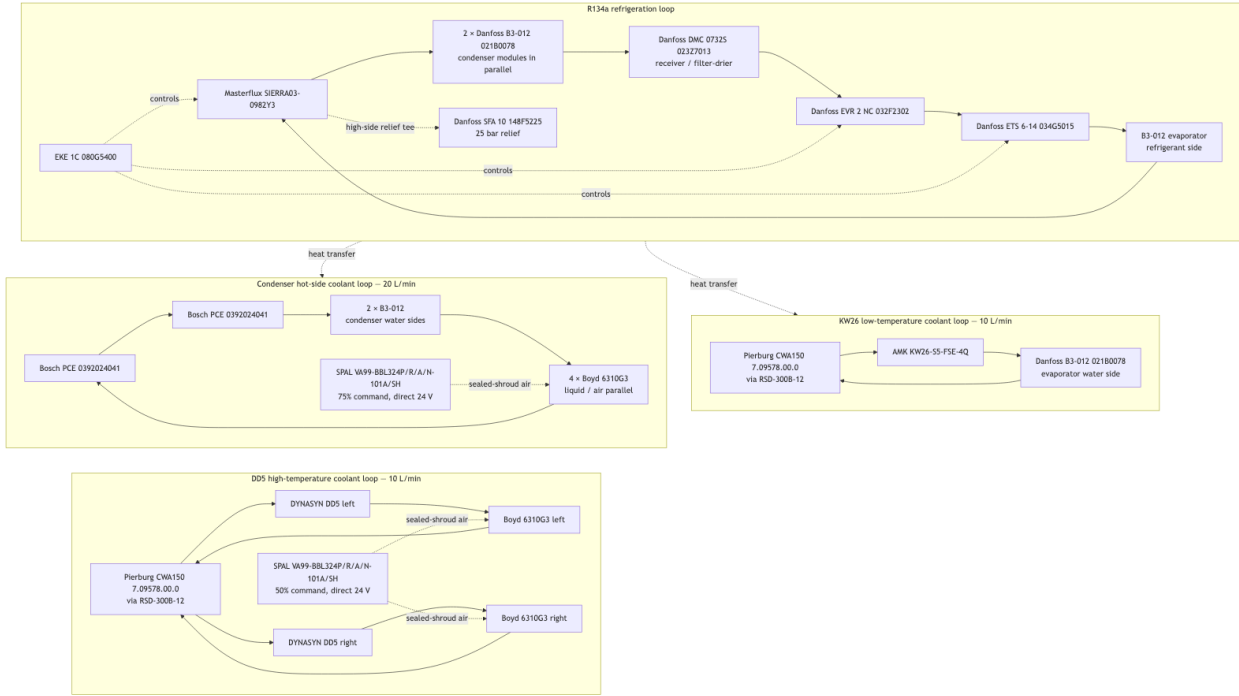


Figure 1: E8 public-data numerical reference topology and component models.

The KW26 and condenser loops have no parallel cooled-component split. The two DD5 branches assume equal 5 L/min flow only for screening; each installed branch must remain above 4 L/min. Final splitter geometry and flow balance require system-curve calculation and bench measurement.

1.1 Active two-temperature architecture

E7 established the governing architecture with two hydraulically independent component temperature levels:

1. a low-temperature loop circulates at least 10 L/min through the KW26 cold plate. A refrigerant evaporator conditions its supply to at most 25°C; the compressor then raises the refrigerant pressure and temperature, and a separate condenser rejects $\dot{Q}_{\text{cond}} = \dot{Q}_{\text{evap}} + P_{\text{compressor}}$

to ambient air;

2. a high-temperature loop circulates through the two user-designed DD5 jackets and passive radiator/fan branches. Each motor branch must retain at least 4 L/min and less than 5 K coolant rise. Operation above the 40°C rated inlet applies AMK's documented motor derating of 1%/K, only up to 60°C; and
3. the two component liquid circuits exchange no coolant. They use separate air shrouds in E8 but share the vehicle electrical and packaging budgets; both heat-rejection paths must be included in installed-duct design.

This separation prevents DD5 heat from automatically increasing refrigeration duty. It does not remove the KW26 control-electronics air-cooling requirement, and sub-ambient surfaces create a dew-point gate. E7 is therefore a replacement engineering envelope, not an FSG requirement and not a selected bill of materials.

E8 instantiates the complete architecture as a public-data numerical model. Its refrigerant condenser uses a secondary water/glycol loop with two parallel B3-012 condenser modules, two series Bosch PCE pumps and four parallel Boyd cores behind one SPAL fan. This loop is separate from both KW26 and DD5 coolant. The refrigerant line adds the exact DMC 0732S receiver/filter-drier and 25 bar SFA 10 relief valve before the EVR/ETS metering hardware. A first CWA150 circulates the KW26 loop through the B3-012 evaporator. A second CWA150 circulates the two parallel DD5 branches through two further parallel Boyd cores behind a second SPAL fan. The two exact 24 V SPAL fans connect directly to the vehicle low-voltage bus; two dedicated DC/DC converters supply only the CWA150 pumps. Exact component identifiers, public sources and every retained derate, conductance, pressure drop, power and mass input are recorded in the E8 research dossier and executable screen. Passing the declared equations establishes numerical feasibility only; it does not select hardware.

2 Competition Compliance Baseline

Status: not a final compliance approval. The legacy rule table mixed revisions and contained unsupported claims. This revision uses Formula Student Germany (FSG) Rules 2026 v1.1 as an initial baseline only. The entered event and its final rule revision must be recorded before any procurement or technical inspection claim is made. The authoritative source is the FSG rulebook¹.

¹Formula Student Germany, [Rules 2026 v1.1](#), checked 9 August 2026.

Table 1: Compliance items to verify against the entered-event rules

FSG reference	Requirement to carry into design	Evidence required before release
T7.2.2–T7.2.9	Cooling medium, driver shielding, sealing, catch-can capacity/-mounting, temperature rating, and discharge routing are regulated. FSG 2026 permits plain water, air, or oil; it does not support the legacy “pure water only” statement.	Event rule revision; schematic; system volume calculation; component temperature ratings; inspection photographs.
EV5.5.16	Tractive-system accumulator cooling/ventilation openings have constraints; cooling architecture must preserve the accumulator enclosure requirements.	Accumulator wall/opening drawing and event-specific compliance review.
EV5.8.3– EV5.8.4	Thermally critical cells require monitoring. The applicable cell limit is 60°C or the cell datasheet limit, whichever is lower.	Cell datasheet, sensor locations, BMS logic, and validation log.
IN9.2	Rain-test requirements affect fan/pump operation and electrical integration.	Event inspection checklist, wiring diagram, and witnessed test result.

No generic fan-power, IP-rating, or noise limit is asserted here: each must be demonstrated by the applicable event handbook and the selected component documentation. The single-loop/multi-loop choice is an engineering trade-off, not a rule conclusion; it remains pending thermal and hydraulic verification.

3 Powertrain Cooling Requirements and Data Status

The vehicle architecture is one AMK KW26-S5-FSE-4Q inverter with two DYNASYN DD5-14-10 motors. The circuit is a candidate single-loop architecture; its final topology, component selections and losses are not approved in this revision.

3.1 Verified manufacturer limits

Table 2: Primary-source cooling constraints

Item	Verified limit and interpretation
AMK inverter	Rated 26 kVA. Its liquid-cooling rated condition is coolant flow temperature $\leq 25^\circ\text{C}$, flow ≥ 10 L/min, pressure 1.5 bar and coolant rise < 5 K. The cold plate can dissipate up to 2,000 W at that condition; this is not an inverter-loss map.
DYNASYN DD5 motor (each)	Rated power 12.3 kW, rated torque 9.8 Nm at 12,000 rpm, maximum torque 21 Nm. The user must design the jacket. Rated cooling requires coolant flow temperature $\leq 40^\circ\text{C}$, flow ≥ 4 L/min and coolant rise < 5 K; derating is 1%/K from 40 to 60°C .
Circuit implication	The inverter's 10 L/min requirement governs total loop flow. With two equal motor branches this nominally gives 5 L/min per motor, but the actual split must be demonstrated by a hydraulic calculation and bench measurement.

Sources are the AMK RACING KIT manual PDK_205481 (sections 6.2.1–6.2.2) and AMK DD5 data sheet A2370DD, retained in the internal evidence dossier.

3.2 Heat-load model: unresolved inputs, not fixed facts

The legacy report used $26 \text{ kVA} \times 2\%$ for inverter loss and 26 kW per motor at 95% efficiency. Neither is an acceptable loss derivation: kVA is not a real-power operating point and the supplied DD5 rated power is 12.3 kW. AMK's archived DD5 workbooks do contain 80/100/120°C total-loss maps by speed and torque; the model now reads those maps directly. For example, the 80°C map interpolates to approximately 1.37 kW total motor loss at 12,000 rpm and 9.8 Nm. The map temperature definition and the fraction of total motor loss transferred to the liquid jacket remain to be established. The previous 3.256 kW continuous and 6.0 kW peak values are therefore retained only as **legacy screening hypotheses**.

For each recorded drive-cycle point, the revised model shall calculate

$$\dot{Q}_{total} = \dot{Q}_{inverter}(V, I, f_{sw}, T_{c,in}, \dot{m}) + 2 \dot{Q}_{motor}(n, \tau, T_{c,in}, \dot{m}), \quad (1)$$

using manufacturer loss maps or measured calorimetry. The DD5 map supplies total motor loss; thermal partitioning to coolant requires a jacket thermal model or test. The inverter loss map remains required. Until these inputs exist, no component is sized from a claimed thermal load.

4 Evidence-Limited Thermal and Hydraulic Baseline

4.1 Purpose and boundary

This section replaces the previous apparent precision in radiator UA , fin area, air velocity and heat-load calculations with a reproducible screening model. The model is stored in `sim/zero_d/` with its inputs and tests. Verified AMK limits remain separate from digitized manufacturer curves and explicitly labelled pump, duct, loss-partition and curve-reading uncertainties.

4.2 Minimum liquid-side capacity

At the inverter-governed minimum flow rate of 10 L/min, using nominal water properties $\rho = 992 \text{ kg/m}^3$ and $c_p = 4180 \text{ J/(kg K)}$, the coolant heat-capacity rate is

$$C_c = \dot{m}c_p = 691 \text{ W/K.} \quad (2)$$

The liquid-side screening heat transport associated with a 5 K rise is therefore

$$\dot{Q}_{5K} = C_c \Delta T = 3.46 \text{ kW.} \quad (3)$$

This is neither radiator capacity nor powertrain heat generation. It is only a necessary water-side check at the stated flow.

Table 3: Legacy-load screening at 10 L/min

Scenario	Legacy heat hypothesis	Coolant rise	5 K screen
Continuous	3.256 kW	4.71 K	passes
Peak, 10 s	6.000 kW	8.68 K	fails

The table must not be read as validation of either heat hypothesis. It demonstrates why transient thermal capacitance, component temperatures and flow distribution must be measured before claiming that a peak event is safe.

4.3 Strict high-ambient design condition

The initial strict screen uses 40°C ambient and no refrigeration. A passive radiator cannot deliver coolant below the ambient temperature in steady operation. It follows directly that the AMK inverter's 25°C rated coolant-inlet condition is **not feasible** in this high-ambient passive case. The prior conclusion that a passive system supports rated operation at 40–45°C is withdrawn.

The internal design study must consequently report distinct operating regimes:

1. rated manufacturer cooling condition (25°C inverter inlet),

2. high-ambient passive operation with the applicable component derating explicitly quantified, and
3. transient peak operation, bounded by measured thermal capacitance and temperature limits.

The internal operating envelope also includes zero-speed heat soak and cold/condensation protection. The 40°C high-ambient scenarios are conservative engineering cases rather than FSG requirements. Each future model or test result shall identify the applicable envelope, source of every boundary condition and hardware revision.

4.4 Governing theory and falsifiable predictions

For a steady liquid loop, conservation of energy gives

$$C_c = \dot{m}c_p, \quad T_{\text{rad,in}} - T_{\text{rad,out}} = \frac{\dot{Q}_{\text{liq}}}{C_c}. \quad (4)$$

The Boyd performance ordinate is a heat-exchanger conductance based on the initial temperature difference, $K_{\text{HX}} = \dot{Q}/(T_{\text{rad,in}} - T_{\text{air,in}})$, rather than an assumed vehicle UA . Consequently,

$$T_{\text{rad,in}} = T_{\text{air,in}} + \frac{\dot{Q}_{\text{liq}}}{K_{\text{HX}}}. \quad (5)$$

For the candidate series order radiator–inverter–parallel motors, the component inlet temperatures are

$$T_{\text{inv,in}} = T_{\text{rad,out}}, \quad T_{\text{mot,in}} = T_{\text{inv,in}} + \frac{\dot{Q}_{\text{inv,liq}}}{C_c}. \quad (6)$$

These equations predict, before any component curve is fitted, that a passive steady loop with positive heat input cannot supply sub-ambient coolant. At 40°C radiator-inlet air it therefore cannot reproduce the KW26 rated 25°C supply condition. Because the KW26 manual also requires the cold plate surface to remain below 40°C, any numerical E3 result with $T_{\text{inv,in}} \geq 40^\circ\text{C}$ disproves that candidate duty point without needing a detailed cold-plate model.

The air-side operating point must satisfy

$$\Delta p_{\text{fan}}(\dot{V}_a, n) = \Delta p_{\text{core+duct}}(\dot{V}_a). \quad (7)$$

Where only the maximum-speed fan curve is published, the fan affinity laws $\dot{V}_a \propto n$ and $\Delta p \propto n^2$ are used only as a labelled sensitivity approximation inside the transformed published curve range. Likewise, the liquid operating point is the intersection of the pump curve and the circuit curve. Quadratic branch-loss sensitivity is written as $\Delta p_i = K_i \dot{V}_i^2$, with equal pressure drop and conservation of flow across parallel branches. A numerical candidate must therefore close energy, air flow, total liquid flow and motor-branch flow simultaneously; passing a stand-alone catalogue-capacity calculation is insufficient.

4.5 Coupled numerical result, not physical validation

The archived DD5 workbook gives a total loss of 1.374 kW per motor at 12,000 rpm, 9.8 Nm and its 80°C map label. The numerical study sweeps DD5 liquid fractions of 0.50, 0.75 and 1.00 and KW26 cold-plate loads of 0.5, 1.0 and 2.0 kW. The selected numerical screen uses two parallel Boyd 6310G3 cores, one SPAL VA99 fan per core at a 0.35 speed ratio, an air-system pressure factor of 1.5, and the CWA150 sensitivity curve. These are calculation boundaries, not installed measurements.

The nominal fan/core intersection is 6.56 m³/min per radiator at 45.6 Pa. Subtracting the recorded 10 Pa fan-curve reading uncertainty moves the intersection below the transformed published fan-curve range, so the air point is not robustly bounded. On the liquid side, subtracting 0.1 bar from the pump curve, adding 0.03 bar to each radiator curve and assuming 1.5 bar non-radiator loss at 10 L/min gives 10.03 L/min total flow. A 2:1 motor-branch resistance ratio leaves 4.16 L/min in the lower-flow branch. The nominal hydraulic requirements are therefore met by only 0.03 L/min total-flow and 0.16 L/min branch-flow margins; this is not a robust approval.

Table 4: Selected E3 coupled thermal screen

DD5 liquid	KW26 heat	Total heat	Coolant rise	KW26 inlet	Result
50%	0.5 kW	1.874 kW	2.70 K	51.9°C	fail
75%	1.0 kW	3.062 kW	4.41 K	59.4°C	fail
100%	2.0 kW	4.749 kW	6.85 K	70.1°C	fail

Every heat-partition combination fails the declared E3 numerical gate. Even the lowest load predicts 54.6°C radiator hot inlet, 51.9°C KW26 coolant inlet and 52.6°C DD5 inlet. The KW26 inlet is already above the manual’s 40°C maximum cold-plate surface temperature, so no detailed cold-plate model can reverse the rejection. The legacy 3.256 kW continuous boundary predicts 65.3°C radiator hot inlet and 60.6°C KW26 inlet. The legacy 6.0 kW/10 s value is retained only as a coolant-domain transient boundary; it is not a component-temperature prediction because solid thermal masses and contact resistances are absent.

The coupled calculation verifies the theoretical high-ambient no-go; it does not physically validate the vehicle. The current Boyd/SPAL/CWA E3 model is therefore rejected within its declared boundary. A replacement numerical candidate must change that infeasible boundary—for example with a lower declared ambient/duty or active refrigeration—and repeat the same coupled checks. Later measurements may calibrate assumptions, but cannot reverse this thermodynamic result at 40°C ambient.

4.6 Finite-volume transient heat-transfer verification

The steady map calculation is independently reproduced with a one-dimensional finite-volume coolant energy equation,

$$\rho V_i c_p \frac{dT_i}{dt} = C_c(T_{i-1} - T_i) + \dot{Q}_i - (UA)_i(T_i - T_{\text{air}}). \quad (8)$$

Upwind advection transports heat around the closed loop; the KW26 and combined DD5 liquid heat are applied at separate cells, and radiator rejection is distributed over the exchanger cells. To preserve the Boyd $K_{\text{HX}} = \dot{Q}/(T_{\text{rad,in}} - T_{\text{air,in}})$ map, the equivalent distributed conductance is

$$UA_{\text{eq}} = -C_c \ln \left(1 - \frac{K_{\text{HX}}}{C_c} \right). \quad (9)$$

The selected run uses 80 cells, 2.0 L assumed loop coolant volume, 10.03 L/min flow, a CFL number of 0.25 and a 0.0374 s nominal time step. It first integrates the 3.062 kW middle sensitivity case for 600 s, then applies the legacy 6.0 kW boundary for 10 s with 2.0 kW assigned to the KW26 liquid path. The baseline converges to 59.45°C KW26 inlet, 60.90°C DD5 inlet and 63.87°C radiator inlet. During the peak, the respective maxima are 61.08°C, 62.43°C and 68.11°C. Sweeping 1.5–3.0 L loop volume changes the peak KW26 inlet from 62.78 to 59.44°C; thermal storage delays the excursion but cannot restore the E3 limit.

Table 5: Finite-volume mesh convergence for the 10 s peak

Cells	Time step	Peak KW26 inlet
20	0.1495 s	61.439°C
40	0.0747 s	61.183°C
80	0.0374 s	61.078°C
160	0.0187 s	61.108°C

The 80–160-cell difference is 0.0294 K and the two finest time-step results differ by 0.0183 K, passing the internal 0.1 K numerical gate. The maximum semidiscrete energy-balance residual is 3.2×10^{-12} W. The finite-volume steady radiator inlet differs from the map equation by 0.078 K. The transient solver therefore verifies energy conservation, storage, advection, radiator rejection and numerical convergence. It is not solid-component conjugate heat transfer; OpenFOAM CHT still requires internal geometry, materials and contact data. The converged coolant-domain result strengthens, rather than replaces, the E3 numerical rejection.

4.7 E7 active two-temperature assumption screen

The E7 screen conditions only the KW26 circuit while retaining a separate passive DD5 circuit. It binds the environmental assumptions to ENV-FSG-SCREEN-v1.0: 40°C and 100 kPa, with 20/50/90% relative humidity and discrete vehicle-speed and stopped-duration qualification points. These

values are an internal screen informed by public FSG and DWD evidence; they are not competition requirements or a recorded drive cycle.

For assumed KW26 cold-plate heat of 0.5–2.0 kW and low-loop environmental heat gain of 0–0.3 kW, the required evaporator range is 0.500–2.300 kW. A deliberately broad COP sensitivity of 1.5–3.0 gives 0.167–1.533 kW compressor input and 0.667–3.833 kW condenser rejection. The separate motor-loop calculation predicts 48.7–57.4°C DD5 inlet over the declared liquid-loss fractions, corresponding to 8.7–17.4% AMK motor derating. Adding the motor radiator and condenser loads gives a shared high-side air-rejection range of 2.041–6.582 kW.

The Masterflux SIERRA03-0716Y3 Rev. E data provide a compressor-only map proxy. At assumed refrigerant $T_e = 12.8^\circ\text{C}$ and $T_c = 50^\circ\text{C}$, its published 24 V points span 1.079–2.312 kW, so one mapped slice reaches the upper E7 load. No mapped slice continuously covers 0.500–2.300 kW, and the public data do not close evaporator/condenser approach, auxiliaries, assembled mass, charge, or low-load control. The result is therefore **numerically plausible as an assumption envelope but blocked for hardware selection**.

At 40°C air the calculated dew points for 20, 50 and 90% RH are 12.8, 27.6 and 38.0°C. A 25°C supply is below dew point in the 50 and 90% cases; insulation or a controlled dry enclosure, drainage, dew-point sensing and fault sequencing are mandatory design inputs. Production CFD remains blocked pending a complete chiller map, installed local air-path CAD and a measured drive cycle. The committed OpenFOAM proxy qualifies only mesh, fan-jump and porous-resistance plumbing against public curves.

4.8 E8 public-data numerical model

Because private vehicle loss and installation data are unavailable, E8 does not stop at the E7 range. It deliberately reuses the legacy 520 W continuous and 2.0 kW peak KW26 liquid-heat values as assumptions, adds 300 W environmental heat gain, and evaluates 0.820 and 2.300 kW evaporator design points. The publicly documented BOM uses a 24 V Masterflux SIERRA03-0982Y3 compressor, three Danfoss B3-012 20-plate heat exchangers (one evaporator and two parallel condenser modules), an ETS 6-14 valve and EKE 1C controller, a Danfoss DMC 0732S receiver/filter-drier, a 25 bar Danfoss SFA 10 relief valve, two CWA150 pumps, two series Bosch PCE pumps, six Boyd 6310 cores, two direct-bus 24 V SPAL fans in separate sealed common shrouds and two pump DC/DC converters. Four Boyd cores reject refrigerant-condenser heat; two form the independent DD5 loop. The relief setpoint is below the 30 bar weakest-component public working-pressure limit.

Relative to the published 35°C compressor table, E8 applies a permanent 0.95 capacity factor and 1.05 input-power factor at 40°C. It derates the B3 design conductance targets by 20%, both pump-head allowances by 10%, and the ideal common-shroud branch airflow to 85%. At 10 L/min, the low loop is limited to an assumed 1.20 bar loss versus 1.26 bar available after derating the public CWA150 25 L/min, 1.4 bar point. The continuous case requires an 89.3% duty fraction at the published 1,800 rpm minimum; the peak case interpolates to 2,929 rpm and approximately 0.680 kW compressor input.

The 75%-speed refrigerant hot-side fan and four-core air-resistance curves intersect at 28.2 m³/min total after a 10 Pa adverse pressure allowance. After the 85% branch-flow factor and digitization penalty, map-bounded conductance is 246.5 W/K. At 2.980 kW condenser rejection, required evaporator/condenser conductances are 168/888 W/K, versus derated design targets of 200/1,200 W/K. The hot-side hydraulic proxy requires 0.569 bar versus 1.08 bar derated series-PCE head.

The independent DD5 loop uses its second CWA150, two parallel 6310 cores and the second direct-bus SPAL fan at 50% speed. An affinity-law transform, 10 Pa adverse fan-pressure allowance, 1.5 bare-core resistance factor and 85% branch-flow factor give 14.7 m³/min total and map-bounded conductance of 125.5 W/K. At the worst retained 2.749 kW coolant-load assumption, the model predicts a 4.36 K coolant rise and 57.5°C DD5 inlet. Its hydraulic proxy is 1.069 bar versus 1.26 bar derated CWA150 head. Conservative wiring input is 1.94–1.97 kW; the fan-affinity operating estimate is 1.48–1.51 kW; mixed verified/assumed mass is 36.0–44.4 kg.

E8 therefore **passes its declared public-data steady numerical gate**. This means that the implemented equations, interpolated public curves and explicit derates are mutually consistent at the stated baseline. It is not physical validation, a hardware-selection recommendation, refrigerant-system safety approval, installed-vehicle CFD or vehicle-release evidence.

4.9 Remaining evidence limits

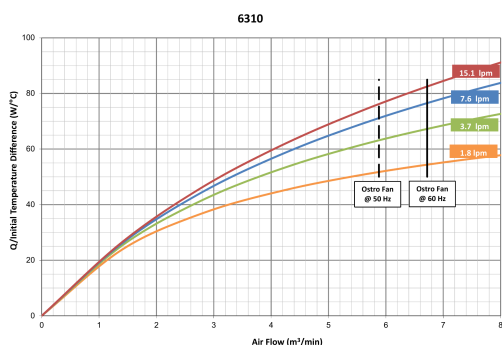
The E3 passive model remains rejected. For E8, the recorded conservative derates bound missing exact-duty B3-012 tables and common-shroud data only as numerical assumptions; they do not replace measurements. Remaining excluded inputs are actual heat-exchanger conductance, complete pump and branch system curves, low-load cycling/bypass dynamics, receiver inventory and charge, relief discharge routing, condensation control, actual KW26/DD5 coolant heat loads, installed CAD and a real drive cycle. Section 12 separates the executed verification from these unresolved physical inputs.

5 Radiator Numerical Screening: E3 Failure and E8 Reference

The former Mishimoto, Setrab and generic motorcycle-radiator conclusions remain withdrawn because no exact heat-rejection and two-sided pressure-drop maps were available. The current study instead evaluated two exact manufacturer-mapped Boyd candidates.

Table 6: Radiator candidate status

Candidate	Model-usable evidence	Decision
Boyd 6210G1	Manufacturer Q /initial-temperature-difference map and exact air/liquid pressure-drop curves; 2.3 kg dry mass and 288 mL volume.	Inferior thermal conductance to 6310G3; retained as a documented comparison, not selected.
Boyd 6310G3	Manufacturer thermal curves at 1.8, 3.7, 7.6 and 15.1 L/min, exact air/liquid pressure-drop curves; 3.9 kg dry mass and 483 mL volume.	Digitized and coupled to the fan, pump and heat-source sensitivity model; fails the declared E3 numerical gate.
Mishimoto MMDB-KX450F-10LX	Exact SKU and basic construction only.	No thermal or pressure-drop maps; packaging candidate only.



(a) Boyd 6310G3 manufacturer graph: Q /initial-temperature-difference versus air flow at four liquid-flow traces. Digitized with a recorded ± 5 W/K reading uncertainty; interpolation only, no extrapolation.



(b) Mishimoto MMDB-KX450F-10LX. Packaging candidate only; no manufacturer thermal or pressure-drop maps were found.

Figure 2: Radiator candidates retained in the numerical record. An image documents identity or a source curve only; it does not imply selection or procurement approval.

For the rejected E3 two-core numerical topology, each 6310G3 receives approximately 5.02 L/min and the nominal fan solution supplies 6.56 m³/min per core. After subtracting the recorded

5 W/K digitization uncertainty from each core, the combined conductance is insufficient to keep the KW26 inlet below 40°C for any declared E3 heat-partition case. The lowest 1.874 kW case predicts 51.9°C at the KW26 inlet. This is a thermodynamic and numerical rejection within the E3 assumptions, not a physical-validation claim or a request for more optimistic curve fitting.

No current radiator is established as a physically validated final competition installation. The E8 numerical reference uses six 6310G3 cores: four reject refrigerant-condenser heat and two cool the independent DD5 loop. Both sealed shrouds apply a 15% branch-flow penalty and the recorded digitization uncertainty. The DD5 fan additionally uses a 50% affinity-law curve transform and 10 Pa adverse pressure allowance inside the published fan/core domains. The complete E8 steady numerical gate passes. This establishes consistency of the assumed network only; installed flow distribution, recirculation, mounts and hose interfaces remain excluded physical inputs.

6 Fan and Air-Path Numerical Screening

The rejected E3 branch used the 12 V SPAL VA99-ABL315P/R/A/N-101A/SH. Its affinity-law sensitivity lost published-curve overlap under the adverse pressure check, while its coupled thermal calculation failed independently.

E8 instead uses two exact SPAL VA99-BBL324P/R/A/N-101A/SH 24 V brushless modules directly from the vehicle low-voltage bus. SPAL's catalogue specifies a 280 mm fan, 16–32 V operating range, PWM input, 1,000–4,000 rpm range, 2.5 kg mass and IP68/IP6K9K protection. Its maximum-speed pressure-flow/current curve is tested at 26 V, $20 \pm 5^\circ\text{C}$ and 1.17 kg/m^3 air density on an ANSI/AMCA 210-compliant bench. Direct 24 V operation deletes two 1.19 kg DC/DC converters from the former package.

The refrigerant hot-side fan runs at a 75% command behind four parallel cores. An affinity-law transform, 10 Pa adverse pressure allowance and 1.5 bare-core resistance factor give $28.2 \text{ m}^3/\text{min}$ and 50.0 Pa. The 85% branch factor gives $5.99 \text{ m}^3/\text{min}/\text{core}$ for heat transfer. The two-module condenser retains 35.2% conductance margin at the peak screen.

The DD5-loop fan runs at a 50% command behind two parallel cores. The same derates give $14.7 \text{ m}^3/\text{min}$, 52.5 Pa and $6.23 \text{ m}^3/\text{min}/\text{core}$ thermal airflow. Worst-case DD5 inlet is 57.5°C , 2.5 K below its 60°C limit.

For wiring, the screen conservatively retains each corresponding maximum-speed-curve current, giving 1.94–1.97 kW package input. The fan-affinity cube gives a 1.48–1.51 kW operating estimate at the commanded speeds; this lower range guides energy control but cannot size wiring until

PWM current is measured. The controller may reduce speed below 75%/50% at lower load only while measured condenser pressure, KW26 supply temperature and DD5 inlet temperature retain their limits.

Both public-map operating points pass the E8 baseline numerical gate. The perturbation matrix in Section 12 shows that the condenser path passes at a 70% fan command and fails at 65%, while the DD5 path passes at 45% and fails at 40%. These are model boundaries, not installed speed/current or shroud-flow validation.

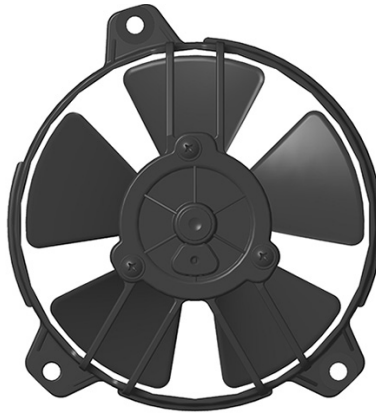


Figure 3: SPAL VA31-A101-46A from the legacy survey. The E8 numerical reference instead uses the 24 V brushless VA99-BBL324P/R/A/N-101A/SH with an exact catalogue curve; the photograph documents the surveyed hardware class only and implies no selection.

7 Pump and Hydraulic Numerical Screening

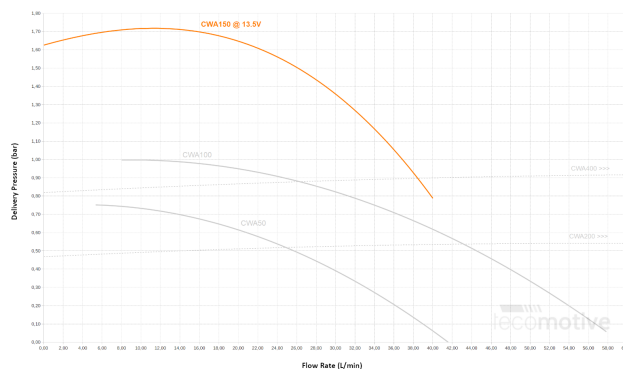
The AMK inverter establishes a minimum 10 L/min flow and documents 1.5 bar working pressure at its rated cooling condition; 1.5 bar is not treated as a required pump differential pressure. Each DD5 branch requires at least 4 L/min. Equal motor branches would receive half the total flow, but the numerical screen also applies a 2:1 branch-resistance imbalance.

The Pierburg CWA150, identifier 7.09578.00.0 / VAG 4KE965567B, remains a sensitivity candidate. Its identity is manufacturer-confirmed, while the complete 13.5 V, 80°C 50/50 water-glycol curve used here is hosted by Tecomotive rather than Pierburg/Motorservice. The model therefore subtracts 0.1 bar from that pump curve. It combines two parallel Boyd 6310G3 liquid curves, adds 0.03 bar reading uncertainty to the radiator branch, and sweeps the unknown remainder of the vehicle system as a quadratic loss referenced at 10 L/min.

At the selected conservative 1.5 bar non-radiator loss, the intersection is 10.03 L/min total at 1.61 bar. With a 2:1 motor-branch resistance ratio, the lower-flow DD5 branch receives 4.16 L/min.

This marginal shared-loop result cannot override the independent E3 thermal no-go; that topology remains rejected.

The independent 20 L/min refrigerant-condenser loop uses two Bosch PCE 0392024041 pumps in series. Each public nominal point is 20 L/min at 0.6 bar; a 10% series-head derate gives 1.08 bar available versus 0.569 bar modelled loop loss. The 70 W public electrical listing is increased by 20% for the upper power budget. All three declared coolant-loop equations pass the E8 baseline numerical gate. A one-pump hot-side perturbation and a 1.30 bar KW26-loop-loss perturbation both fail, exposing the model boundary. Installed pressure/flow behavior remains outside the verified numerical input.



(b) Tecomotive-hosted pressure-flow curve (secondary source; the model applies a -0.1 bar adverse shift). A first-party exact-part curve remains an open supplier-data request.

17

8 Overflow Containment and Catch Can

FSG 2026 T7.2.2–T7.2.9 is the current compliance baseline. The final catch-can volume, mounting, materials, discharge routing and system sealing must be checked against the entered-event rule revision and the measured filled-system volume. No commercial catch can is selected in this revision.

The required engineering calculation is based on the actual cold fill volume, maximum coolant temperature, coolant expansion, trapped volume and pressure-cap/overflow arrangement. The resulting vessel and hose routing are then inspected on the built car. The former fixed total-loop volume and product safety factors are withdrawn because the radiator, hose lengths and final topology are not yet fixed.



Figure 5: Canton Racing Products 80-251 (nominally 443 mL) from the legacy survey. Retained as a surveyed reference only: volume, mounting, materials and discharge routing remain unselected until the measured filled-system volume and final topology exist.

9 Degassing and Header-Tank Strategy

A header tank or swirl pot may improve filling, deaeration and pump-inlet robustness, but it is not demonstrated as an absolute requirement by the present evidence. Its need and geometry depend on the final loop high points, radiator arrangement, pump NPSH requirement and observed air release during filling and operation.

The selection test is practical: fill and bleed the assembled loop, repeat heat-up/cool-down cycles, observe pump cavitation/flow stability and inspect retained air at all high points. A selected vessel must also satisfy final volume, mounting, material and FSG overflow requirements. Product-specific centrifugal-separation and capacity claims are deferred until their geometry and test evidence are available.



Figure 6: Radium Engineering 750 mL swirl pot from the legacy survey. Retained as a surveyed reference only: its necessity, capacity and separation performance are undemonstrated by the current evidence.

10 Coolant Chemistry, Hoses and Fittings

The legacy statement that FSAE universally permits only deionized water and prohibits additives is removed. FSG 2026 permits plain water, air or oil in T7.2.2, while the AMK manual specifies circulating-water quality guidance (pH 7–9, conductivity limit) and calls for a corrosion inhibitor; it also warns against electrolysis when more noble materials, such as copper, are mixed with its aluminum alloys. The entered-event rules and all component manufacturers must be reconciled before coolant is selected.

No AN size, hose, fitting or adapter is selected in this revision. The final specification must demonstrate: pressure and temperature rating; compatibility with the final coolant and aluminum components; bend radius and routing; G1/4 interface integrity; vibration retention; electrical isolation where required; and measured contribution to the hydraulic system curve. Material choice follows the verified component interfaces and corrosion-control plan rather than generic motorsport preference.



Figure 7: Goodridge 811-series hose from the legacy survey, shown as a representative motorsport product family. No hose, AN size or fitting is selected; the final specification must satisfy the interface, compatibility and measured-loss requirements above.

11 Future Physical Validation Plan

Not executed in Revision 12. The present deliverable is restricted to numerical verification, public-evidence closure and executable data-admission contracts. The following future measurements would calibrate or challenge its assumptions; they are not evidence used to obtain the reported numerical results.

1. **Hydraulic characterization:** measure pressure/flow curves for the KW26, DD5 and refrigerant-condenser coolant loops, including each DD5 branch and every radiator branch, using final hardware and coolant chemistry. Compare each measured system curve to its CWA150 or series-PCE pump bound.
2. **Radiator and fan calorimetry:** characterize both sealed shrouds with traceable heater loads; measure every branch flow, coolant and air inlet/outlet temperatures, air-side differential pressure, recirculation and fan electrical power over the operating range.
3. **Thermal transient:** apply a representative, recorded heat-load sequence and measure component-simulator, coolant and air temperatures. Use the result to identify thermal capacitance and peak-temperature margin.
4. **Fill, bleed and overflow:** perform repeated fill/bleed and heat/cool cycles. Record trapped-air observations, pump-flow stability, overflow volume and leak inspection.
5. **Active-loop calorimetry and control:** map evaporator capacity, compressor and auxiliary input, condenser rejection and low-load control over the intended refrigerant saturation-temperature and coolant-flow envelope. Verify $\dot{Q}_{\text{cond}} = \dot{Q}_{\text{evap}} + P_{\text{input}}$ within instrumentation uncertainty.
6. **Refrigerant integrity and protection:** a qualified refrigeration technician shall verify every pressure-containing part, joint, pressure test, evacuation record, measured charge, receiver fill limit, leak result and SFA 10 discharge route before the first energized thermal test. The 25 bar relief hardware is not a substitute for this gate.
7. **Condensation and fault response:** repeat the low-temperature operating sequence across the approved humidity/dew-point envelope. Verify insulation or enclosure performance, drainage, sensor trip thresholds and safe response to pump, compressor and fan loss.

The first acceptance criterion is evidence completeness: every sensor has a calibration/accuracy record and every test has a configuration, ambient condition and raw-data archive. The thermal energy-balance residual target is within 5% after instrumentation uncertainty is quantified. Component limits (flow, temperature, pressure and coolant rise) must be met under the conditions to which their manufacturer specifications apply.

Two targeted component tests close the present powertrain-data gaps. Inverter calorimetry shall synchronize HV DC input, AC output, 24 V control input, cold-plate coolant heat and absolute inlet/outlet pressure at each defined hardware, firmware and control point. Motor testing shall compare electrical input, shaft power, AMK map total loss and actual-jacket coolant heat at defined speed/torque points. The required equations, instrumentation records and acceptance gates are defined in the internal component-loss characterization procedure. Reviewed steady points must also pass `sim/zero_d/calorimetry_gate.py`; version 2 requires distinct operator and reviewer, raw-file hashes, complete calibration provenance and the declared energy-balance closure. Accepted results remain point-limited and do not authorize interpolation or extrapolation. The supplier request matrix separately defines the exact AMK, Pierburg, Danfoss, Boyd/SPAL and active-stage documents that may replace measurements. A component identity, family maximum, separate maximum flow/pressure pair or unconditioned heat-exchanger range never satisfies that request.

12 Executed Numerical Verification

12.1 Scope, provenance and reproduction

Revision 12 makes the reduced-order numerical record reproducible from one command:

```
python3 sim/zero_d/numerical_campaign.py
```

The runner loads the versioned E3, E7 and E8 inputs; executes every reduced-order model; resolves numerical transitions; runs five deterministic interaction grids and independent checks; then writes a JSON result, six CSV tables, five engineering plots and a SHA-256 manifest into the ignored `generated` directory. The manifest records digests for every runtime input—including the archived DD5 loss-map bundle—every model and generated artifact, the Git revision, the declared analysis domains and the interpolation-only curve policy. The build command runs both test suites and rebuilds the generated files before compiling the report.

The reduced-order and measured-data admission suite contains 59 tests; the OpenFOAM method-qualification suite contains 21 tests. The executed model hierarchy is:

1. the E3 algebraic network solves fan/core and pump/system intersections, coolant heat-capacity rate and manufacturer radiator conductance;
2. an independent E3 one-dimensional finite-volume model verifies advection, thermal storage, distributed heat rejection, conservation and mesh/time-step convergence;
3. the E7 model sweeps the active two-temperature energy envelope, compressor COP, ambient

- heat leak, humidity and DD5 derating;
4. the E8 model couples public compressor, heat-exchanger, fan, radiator and pump inputs in a steady three-liquid-loop calculation;
 5. exact one-variable boundaries and five-by-five interaction grids expose the local E8 pass/fail domain without assigning probability distributions; and
 6. a remote OpenFOAM 14 surrogate-duct campaign exercises a tabulated fan pressure jump, a fitted porous-core loss, conservation monitoring and systematic four-mesh sensitivity without representing the installed car.

12.2 Public-evidence closure boundary

The 15 August 2026 primary-source extension rechecked the decision-critical KW26, DD5, CWA150, Danfoss B3-012 and FSG duty inputs. It verifies exact identities and narrows documented limits, but it supplies no condition-complete application KW26 loss, DD5 jacket-heat fraction, exact-part CWA150 pressure-flow/electrical/control map, B3 evaporator or condenser duty, or target-vehicle power waveform. Public course dimensions and timing results are test-envelope facts, not a synthetic drive trace. Separate pump maximum flow and maximum pressure are not a paired operating point; B3 family area or capacity is not an exact UA or pressure drop. Consequently E8 retains its original numerical assumptions and receives no hardware-status upgrade. Controlled supplier requests and machine-checked KW26/DD5/vehicle-data admission contracts define the next executable closure.

12.3 E3 independent verification

The steady E3 model fails all nine declared KW26/DD5 heat-partition cases. Its lowest 1.874 kW case gives a 51.9°C KW26 inlet at 40°C radiator-inlet air; the middle 3.062 kW case gives 59.4°C. The finite-volume model independently converges to 59.45°C for that middle baseline and reaches 61.08°C during the 10 s, 6.0 kW boundary. Figure 8 shows that all three numerical routes remain above the manually imposed 40°C cold-plate boundary.

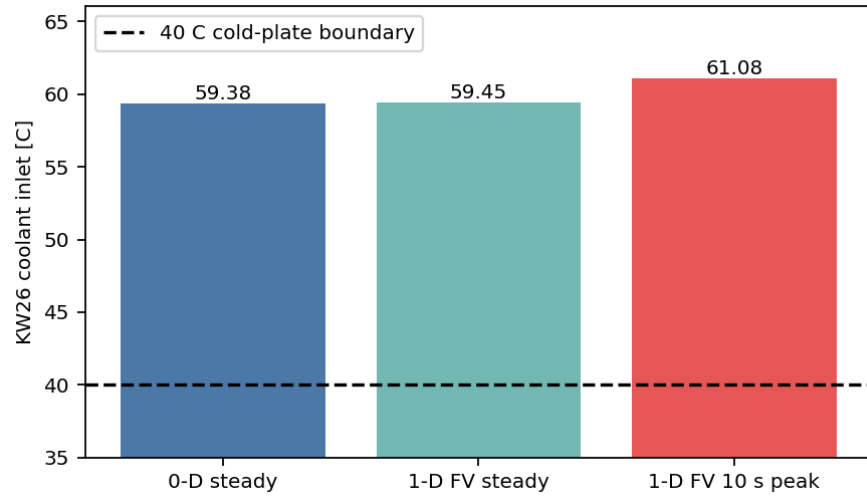


Figure 8: E3 algebraic and finite-volume KW26 inlet results. The agreement checks implementation consistency; the common failure is the engineering result.

The 80–160-cell difference is 0.0294 K, the finest time-step difference is 0.0183 K and the steady map/FV mismatch is 0.0777 K. The maximum semidiscrete energy residual over the baseline, peak and recovery stages is 3.183×10^{-12} W, below the 10^{-8} W campaign gate. Figure 9 reports the two refinement sequences. The model therefore passes conservation, convergence, steady/transient consistency, monotonic-response and curve-domain checks while its E3 thermal gate fails.

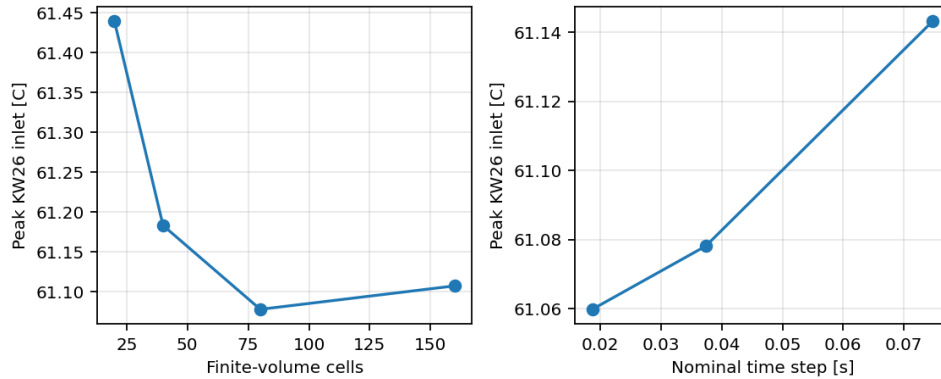


Figure 9: E3 finite-volume mesh and time-step convergence for peak KW26 inlet temperature.

12.4 E7 and E8 baseline results

E7 remains an assumption-envelope result: 0.500–2.300 kW evaporator load, 0.167–1.533 kW compressor input, 48.7–57.4°C DD5 inlet and 2.041–6.582 kW combined high-side rejection. It is numerically plausible, but its broad COP and heat-partition sweeps are not a component model.

The E8 public-data model passes both declared steady design points:

Table 7: E8 baseline steady numerical result

Case	Q_e	Speed	KW26 rise	UA_e req.	UA_c req.	Gate
Continuous	0.820 kW	1,800 rpm	0.83 K	39 W/K	152 W/K	pass
Peak	2.300 kW	2,929 rpm	3.17 K	168 W/K	888 W/K	pass

At baseline, the refrigerant hot-side model gives 28.2 m³/min airflow, 246.5 W/K conservative radiator conductance and 35.2% peak condenser-UA margin. The DD5 path gives 14.7 m³/min, 125.5 W/K and a worst assumed 57.5°C inlet at 2.749 kW coolant heat. The conservative 24 V input range is 1.94–1.97 kW; the fan-affinity operating estimate is 1.48–1.51 kW. These are equation outputs under declared assumptions, not measured performance.

12.5 Resolved E8 boundaries

Bisection was restricted to declared input domains and terminated only when a pass-side and fail-side point were separated by the stated tolerance. Table 8 reports the pass-side values; full precision and the paired fail-side values are retained in `exact_boundaries.csv`.

Table 8: Resolved one-variable E8 numerical boundaries

Boundary	Baseline	Pass side	Limiting gate
Minimum condenser-fan command	0.750	0.680168	condenser UA margin
Minimum DD5-fan command	0.500	0.435217	DD5 inlet below 60°C
Minimum common B3 usable- UA factor	0.800	0.670258	evaporator UA margin
Maximum KW26-loop pressure loss	1.200 bar	1.259998 bar	available pump head
Maximum two-motor DD5 coolant heat	2.749 kW	3.13434 kW	DD5 inlet below 60°C

These values correspond to only 9.3% condenser fan reduction, 13.0% DD5 fan reduction, 16.2% common B3 usable- UA reduction, 5.0% KW26-loop pressure-loss increase and 14.0% DD5 coolant-heat increase from baseline. Figure 10 normalizes each transition by its baseline value. These local margins are too narrow to absorb unknown installed duct, tolerance and exact-duty component effects by assertion.

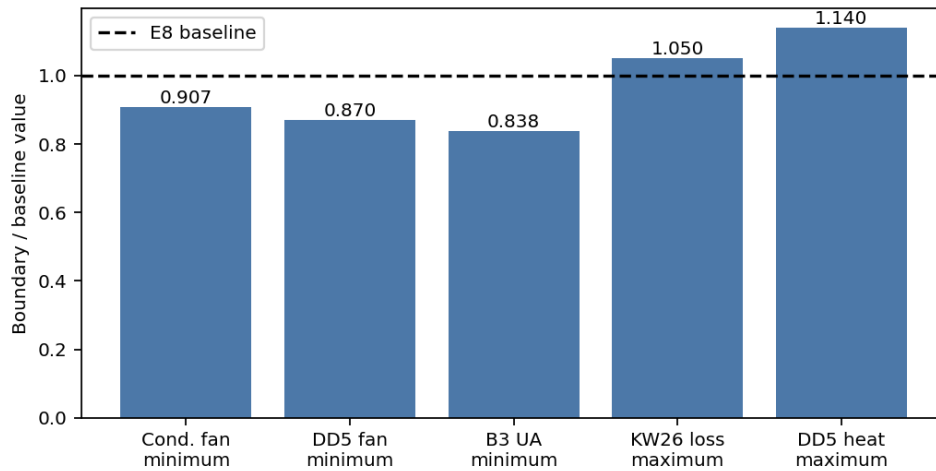


Figure 10: Resolved E8 boundaries normalized by their corresponding baseline inputs. Values below one are minimum retained capability; values above one are maximum tolerated demand.

The ambient/KW26-heat sweep spans 5–40°C and 0.500–2.000 kW. Every sampled point passes the declared E8 equations, so no transition is reported inside that domain. The implementation does not extrapolate beyond 2.000 kW to manufacture a boundary.

12.6 E8 interaction domains

Five 25-point grids vary two inputs at a time: condenser-fan command against B3 usable UA ; DD5-fan command against motor-loss liquid fraction; ambient temperature against KW26 heat; KW26-loop loss against available pump head; and worst shroud branch flow against usable radiator conductance. The first, second, fourth and fifth grids contain both pass and fail points. The ambient/heat grid remains entirely inside the pass domain.

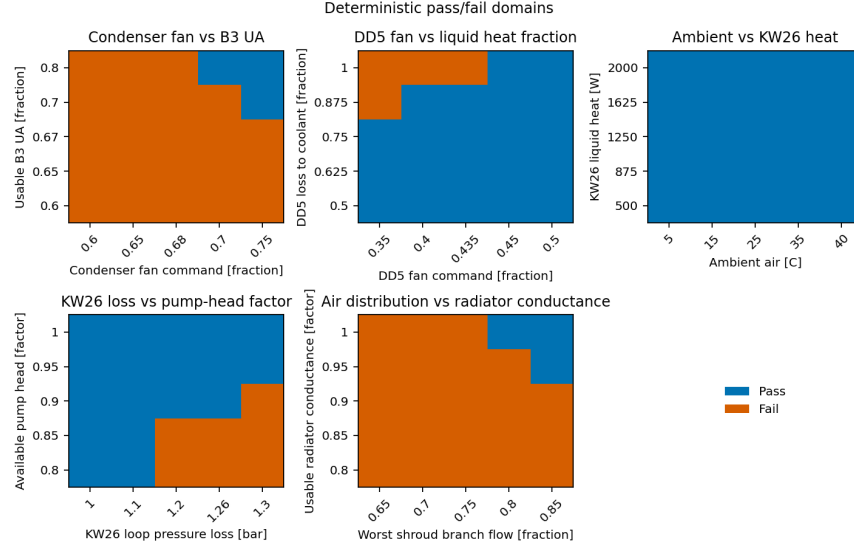


Figure 11: Deterministic E8 two-parameter pass/fail grids. Each cell is a direct model execution; the stepped edges are sampling boundaries, not fitted probability contours.

The interaction grids expose coupled loss of margin. For example, 70% condenser-fan command passes with the baseline 0.80 B3 factor but fails at 0.70, while 75% fan command still passes at 0.70. Similarly, increased KW26-loop loss becomes unacceptable as available pump head is derated. Therefore the one-variable limits in Table 8 must not be combined independently as a new design point.

12.7 OpenFOAM status and excluded physics

The offline preflight fits the public radiator pressure curve with a maximum residual of 2.571 Pa and RMS residual of 1.494 Pa. A reviewed remote OpenFOAM 14 build executed a tutorial smoke test and 7,440/53,940/431,520/3,481,920-cell coarse/medium/fine/extra-fine surrogate-duct cases. All four meshes pass checkMesh, remain within the transformed $0.1069\text{--}0.2382\text{ m}^3/\text{s}$ fan curve, satisfy the 0.5% global mass-balance gate and settle below the 1% final-window drift gate.

The original three-level campaign failed because medium-to-fine mass flow and pressure loss changed by 3.169% and 3.081%. Before the extension ran, the same 2% threshold was retained and a systematic extra-fine level was added. Fine-to-extra-fine changes are 0.959% for mass flow and 0.0145% for pressure loss, so the extended surrogate campaign passes. The accepted qualification point is $0.129484\text{ m}^3/\text{s}$ (0.144048 kg/s) and 36.078 Pa. The medium/fine/extra-fine mass-flow sequence gives observed order 1.741 and finest-grid GCI 0.508%. The nearly unchanged pressure result gives a formal order of 7.730 and GCI 0.000084%; this cancellation is reported as a numerical

diagnostic, not physical uncertainty.

The constant core-matched duct and ideal full-face actuator deliberately exclude installed fan-footprint, shroud, sidepod and wake maldistribution; no thermal field is solved.

The steady, zero-dimensional E8 calculation additionally excludes compressor cycling, refrigerant inventory and oil return, solid temperatures and contact resistance, local boiling and cavitation, recirculation, three-dimensional duct losses, and structural/electrical integration. Exact-duty B3 tables, complete first-party CWA pump curves, measured KW26/DD5 coolant heat partitions and a real drive cycle are also absent. Revision 12 therefore verifies the reduced-order equations and declared assumption boundaries and qualifies the surrogate air-path discretization within its stated gates. It does not qualify installed-vehicle CFD and makes no hardware, physical-validation or vehicle-release claim.

12.8 Numerical decision

1. Reject the passive E3 Boyd/SPAL/CWA configuration for the declared 40°C-air envelope; increasing passive radiator detail cannot restore the KW26 25°C rated inlet below ambient.
2. Retain the separated, actively conditioned E8 architecture only as the current numerical reference. Its baseline passes, but its coupled local margins do not authorize a component purchase.
3. Accept the four-level grid qualification only for the idealized core-matched surrogate. Do not transfer its operating point or convergence uncertainty to an installed vehicle.
4. Admit a production design decision only after controlled exact-part supplier data, accepted KW26/DD5 and installed hydraulic measurements, a real vehicle drive cycle, controlled installed air-path CAD and boundary data enter a revised, grid-qualified model and the physical validation plan is executed.

13 Consolidated Candidate Hardware Register

This register consolidates every component that appears in the E8 numerical reference configuration. It is a traceability table, **not a purchase order**: no row below is approved for procurement. Three gates must pass before any row becomes a purchase line:

1. **Measured heat loads.** KW26 and DD5 bench calorimetry accepted by `sim/zero_d/calorimetry_gate.py` replaces the assumed 0.500–2.000 kW KW26 heat range and the 0.5–1.0 DD5 liquid-fraction sweep.

2. **Supplier data.** The exact-part documents listed in docs/validation/supplier-data-request-matrix.md replace secondary curves and assumed heat-exchanger duties.
3. **AMK derating answer.** If AMK confirms in writing a usable continuous operating region above the 25 °C rated coolant condition, the refrigeration rows (compressor, plate heat exchangers, valve, receiver, relief) may be deleted entirely and the register reverts to a passive configuration screened at the realistic Hockenheim weather envelope.

Table 9: E8 candidate hardware register — numerical reference only

Role	Candidate model	Verified data	Status and purchase blocker
KW26-loop pump	Pierburg CWA150, 7.09578.00.0 / VAG 4KE965567B	Exact identity first-party; 12 V family page.	Candidate. Curve is secondary (Tecomotive); needs first-party curve and installed loop bench map.
DD5-loop pump	Pierburg CWA150 (second unit)	As above.	As above.
Condenser-loop pumps	2× Bosch PCE 0392024041 in series	24 V, 20 L/min at 0.6 bar nominal point, PWM, IPX7, 0.665 kg, 50/50 water-glycol.	Candidate. Nominal point only; needs full curve and series bench verification.
Condenser cores (×4)	Boyd 6310G3	Manufacturer thermal and pressure maps digitized; 3.9 kg, 483 mL, 10.3 bar test pressure.	Candidate. Needs machine-readable vendor tables and common-shroud bench test.
DD5-loop cores (×2)	Boyd 6310G3	As above.	As above.
Fans (×2)	SPAL VA99-BBL324P/R/A/N-101A/SH	24 V brushless, 16–32 V, PWM, 1,000–4,000 rpm, 2.5 kg, IP68/IP6K9K; 26 V bench curve.	Candidate. Needs installed PWM current and shroud airflow measurement.
continued on next page			

Table 9: E8 candidate hardware register (continued)

Role	Candidate model	Verified data	Status and purchase blocker
Compressor	Masterflux SIERRA03-0982Y3 (high capacity)	24 V tables, 1,800–4,200 rpm, R134a/R513A/R1234yf, 54.4 °C condensing rating condition.	Candidate. Compressor table only; needs a complete-stage map at 40 °C ambient or derating via the derating branch.
Evaporator	Danfoss B3-012, 20 plates, 021B0078	Exact identity, materials, ports and working envelope; 1.312 kg.	Candidate. Duty, UA , both-side pressure drops and approach are assumptions pending a condition-complete HEXSelector output.
Condenser heat exchangers	2× Danfoss B3-012 021B0078 parallel	As above.	As above.
Expansion valve + controls	Danfoss ETS 6-14 034G5015, coil controller 034G5115, EKE 1C 080G5400	4.5 kW R134a nominal, 10–100% capacity-control range.	Candidate contingent on the refrigeration branch surviving gate 3.
Receiver/filter-drier	Danfoss DMC 0732S 023Z7013	6.3 kW R134a flow capacity, 42 bar maximum working pressure, 0.165 L.	Verified ratings; same contingency.
Relief valve	Danfoss SFA 10 148F5225	25 bar set pressure, 65 bar maximum working pressure.	Verified ratings; discharge routing and technician sign-off still required.
Catch can	None	—	Blocked: requires measured filled-system volume and final topology.
Swirl pot / header tank	None	—	Blocked: necessity undemonstrated; decided by fill/bleed testing.

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Table 9: E8 candidate hardware register (continued)

Role	Candidate model	Verified data	Status and purchase blocker
Hoses, fittings, sizes	None	—	Blocked: requires final interfaces, coolant chemistry and measured system curve.
Coolant chemistry	None	—	Blocked: FSG T7.2.2, AMK water-quality and inhibitor guidance, and copper/aluminium materials must be reconciled.

Two consequences follow. First, the register shows why no purchase list can be issued today: every performance-critical row carries an open evidence blocker, and the refrigeration half of the table may be deleted outright by the AMK derating answer. Second, the zero-purchase path is fully specified: the inverter and motors are already owned, so bench calorimetry (Section 11) and the supplier requests close the blockers without buying any cooling hardware.